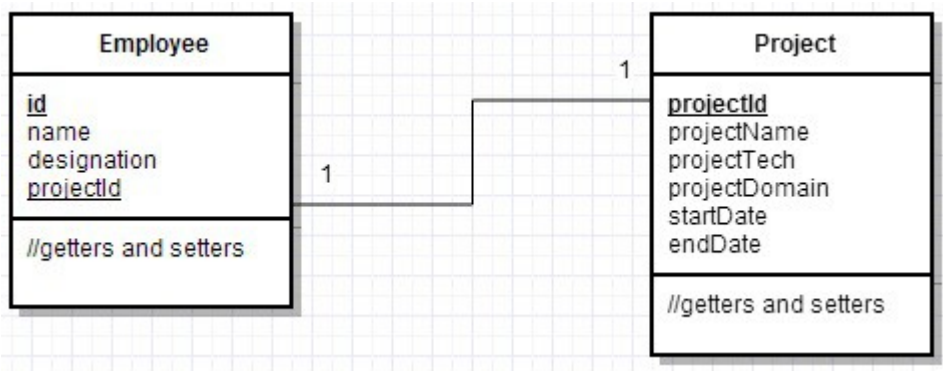
HIBERNATE PRACTICE QUESTIONS

1. Create SessionHelper class in a separate helper package and do following

- public methods to *Session getHibernateSession()* and *closeHibernateSession(Session session)*
- Create an object of Configuration class
- Initialise/Instantiate ServiceRegistryBuilder by applying default settings of configuration generally mapped to default hibernate.cfg.xml file
- Build session factory by using sessionfactory inerface and open up valid Hibernate Session
- Close existing session in current hibernate application.
- Test Run Main Method and try to print out valid session object.

2. One-to-One(Use One-to-One / Many-to-one tags)

    - Create a class Employee and Project(as shown below in the image)



    - Create Employee.hbm.xml & Project.hbm.xml files
    - Create hibernate.cfg.xml with valid mappings of Employee & Project POJOs
    - Map One Employee to One Project using one-to-one XML tag in Employee.hbm.xml
    - Create Test Main Class to insert few records in employee and project table(as shown)

ID	NAME	DESIGNATION	PROJECTID
566,568	SUNNY JOSHI	ITA	1,015,557
566,580	RAHUL PANDEY	ASE	1,015,558
566,591	SWETA ARORA	ASSISTANT CONSULTANT	1,015,560
566,593	SHIFA SADDI	CONSULTANT	1,015,561
566,598	AJAY SHARMA	PRINCIPAL CONSULTANT	1,015,562

PROJECTID	PROJECTNAME	PROJECTTECH	PROJECTDOMAIN	STARTDATE	ENDDATE
1,015,557	ECOMDEV	JAVA	RETAIL	5/28/2014 12:00:00 AM	5/28/2015 12:00:00 AM
1,015,558	ECOMTEST	SQL	RETAIL	5/5/2014 12:00:00 AM	5/5/2016 12:00:00 AM
1,015,560	PRODSUPPORT	.NET	INSURANCE	7/5/2014 12:00:00 AM	12/5/2016 12:00:00 AM
1,015,561	MONITORING	C++	TELECOM	7/5/2015 12:00:00 AM	12/5/2018 12:00:00 AM
1,015,562	MIGRATION	UNIX	BFS	7/5/2014 12:00:00 AM	12/5/2019 12:00:00 AM

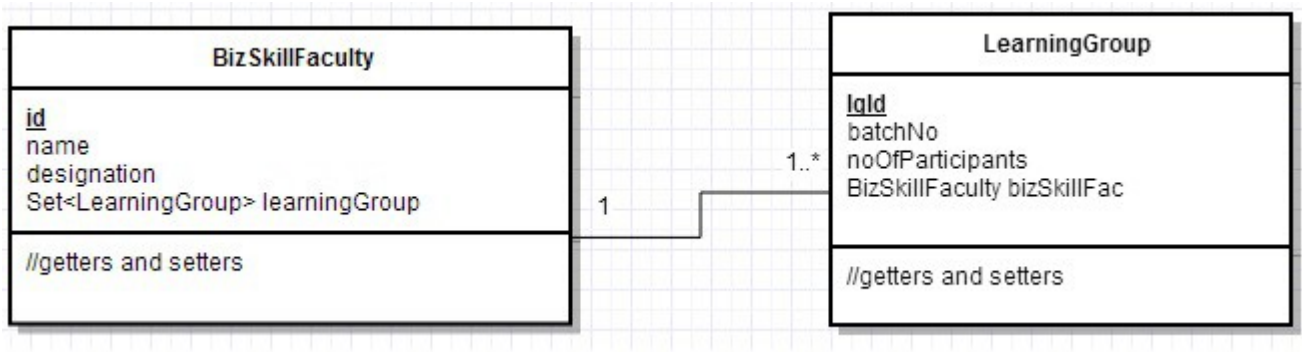
HQL on the same table

- Implement from clause in the above and fetch all records of employee
- Iterate over each record and display relevant details of each employee in eclipse console.
- Display project name, technology, start date, end date from Project Table(use : Select clause)
- Display project id, project name, technology, start date, end date whose project domain is “RETAIL” (use : where clause)

Optional : Try the same case study using Annotations

3. One-to-Many(Use One-to-Many tag)

- Create a class BizSkillFaculty(as shown below in the image)
- Create a class LearningGroup(as shown below in the image)
- Create BizSkillFaculty.hbm.xml & LearningGroup.hbm.xml files
- Create hibernate.cfg.xml with valid mappings of BizSkillFaculty & LearningGroup POJOs
- Map One BizSkillFaculty to Many LearningGroup using one-to-many XML tag in BizSkillFaculty.hbm.xml
- Create Test Main Class to insert few records in BizSkillFaculty and LearningGroups table(as shown)



BIZSKILLID	NAME	DESIGNATION
576,878	SHARMILA	EXECUTIVE
576,876	NITYANAND	CONSULTANT
576,880	SARITA	EXECUTIVE
576,899	ANUPAM	CONSULTANT

LGID	BATCHNO	NOOFPARTICIPANTS	BIZSKILLID
A56	AHD16	40	576,878
A96	AHD19	40	576,878
A58	AHD16	40	576,876
A60	AHD16	40	576,876
A92	AHD19	40	576,880
A91	AHD19	40	576,880
A93	AHD19	40	576,878

HQL on the same table

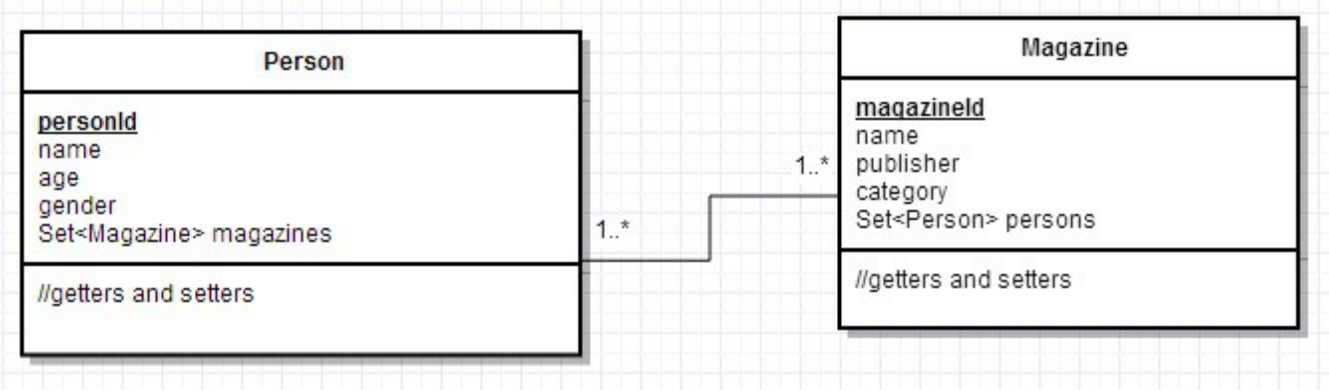
- Implement pagination to fetch Max 2 lg results of 'SHARMILA'
- Implement named query to fetch all Learning Group Details assigned to 'NITYANAND'
- Try out different fetching mechanism in the same case study(***join/select/subselect***)
- Display Bizskills lead name,id, designation who are assigned to batch number 'AHD16'

Optional : Try the same case study using Annotations

4. Many-to-Many(Use Many-to-Many tag)
    - Create a class Person(as shown below in the image)
    - Create a class Magazine(as shown below in the image)
    - Create Person.hbm.xml & Magazine.hbm.xml files
    - Create hibernate.cfg.xml with valid mappings of Person & Magazine POJOs
    - Map Many Person to Many Magazine using many-to-many XML tag in Person.hbm.xml and Magazine.hbm.xml(*try out inverse=true/false*)
    - Create Test Main Class to insert few records in Person and Magazine table(as shown)
    - Basic concept : A person can read many magazines/A magazine can be read by many

persons

- Create a glue table Person_Magazine to join both Person and Magazine tables(as shown)



PERSONID	NAME	AGE	GENDER
101,991	SANJAY	34	M
101,990	RAHUL	32	M
101,992	SWETA	32	F
101,994	JOSEPH	54	M
101,996	SAVITA	54	F

MAGAZINEID	NAME	PUBLISHER	CATEGORY
200,014	INDIA TODAY	AROON PURIE	CURRENT AFFAIRS
200,016	DIGIT	9.9 MEDIA	TECHNICAL
200,018	PC	ZIFF DAVIS	COMPUTER WORLD
200,020	CSR	CSR	GENERAL KNOWLEDGE
200,022	STARDUST	NARI HIRA	BOLLYWOOD
200,024	AUTOMOBILE	ERIC SCHWAB	AUTOMOBILES

PERSONID	MAGAZINEID
101,990	200,014
101,990	200,018
101,990	200,024
101,991	200,014
101,992	200,018
101,992	200,022
101,994	200,014

HQL on the same table

- Find no. of persons who are reading 'INDIA TODAY' magazine(*use: joins/Aggregate*)
- Display list of names who are reading more than one magazine(*use: joins/Aggregate*)

- Display list of magazines which is being read by more than 1 person.
- Display list of name who are reading 'PC'